

MYZ307E

Machine Learning for EE

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TERM PROJECT

Power System Fault Detection & Classification

Using Ensemble Machine Learning Methods

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Problem Definition & Motivation

Why this matters — and why ML?

The Problem

- Transmission line faults (LG, LL, LLG, LLL, LLLG) cause major outages if undetected within milliseconds.
- Traditional relay protection uses fixed thresholds — not adaptive to modern smart grid conditions.
- 3-phase current & voltage signals carry learnable fault signatures that ML can exploit without manual tuning.

Two-Stage Approach

Stage 1 — Detection

Binary: Normal vs. Fault

Dataset: 12,001 samples | 54.2% Normal / 45.8% Fault



Stage 2 — Classification

6-class: Which fault type?

Dataset: 7,861 samples | 6 fault classes

No
Fault

LG

LL

LLG

LLL

LLL
G

Reference Paper & GitHub Repositories

What we built upon — and their limitations

Primary Reference Paper

T. Anwar, C. Mu, M. Z. Yousaf et al., "Robust Fault Detection and Classification via Ensemble ML Models" — Scientific Reports, Nature, 2025

DOI: 10.1038/s41598-025-86554-2 | RF-LSTM Tuned KNN ensemble → 99.96% accuracy

1 Repo 1 — svdeepak99

github.com/svdeepak99/Machine_Learning-Power_System_Fault_Detection

- SVM, Decision Tree, RF, PCA, GMM, MLP
- 12,001 samples — binary detection only

⚠ No classification — no ensemble

2 Repo 2 — KingArthur000

github.com/KingArthur000/Electrical-Fault-detection-and-classification

- MATLAB Simulink dataset, 7,861 valid samples
- Detection + classification attempted

⚠ Poly. regression for classification — inappropriate

Dataset & Preprocessing

Real values from notebook output

Features ($x \in \mathbb{R}^6$)

Ia Phase A Current

Ib Phase B Current

Ic Phase C Current

Va Phase A Voltage

Vb Phase B Voltage

Vc Phase C Voltage

Dataset 1 — Detection (powerline_dataset.csv)

12,001 samples • Binary: 0=Normal, 1=Fault • 6 features



Dataset 2 — Classification (classData.csv)

7,861 valid samples • 6 features • Multi-label [G,C,B,A] → 6 classes



⚙ StandardScaler (fit on train only) • 80/20 stratified split • PCA: 4 components → 96.9% variance

Contributions (1/2) — Pipeline & Model Expansion

Baseline gaps and our solutions

✗ Baseline Gap

Repo 1: binary detection only —
no fault type classification



✓ Our Solution

Added 6-class classification stage using Dataset 2.
Unified two-stage pipeline: Detection → Classification.

✗ Baseline Gap

Repo 1 evaluates 5 models only.
Repo 2 uses polynomial regression
for classification — inappropriate.



✓ Our Solution

Evaluated 8 classifiers on both tasks:
+KNN, Naive Bayes, Logistic Reg., Gradient Boosting.
Replaced poly. regression with proper classifiers.

✗ Baseline Gap

No ensemble method in
either repository.



✓ Our Solution

VotingClassifier(RF + SVM + GB, voting='soft')
for both detection and classification tasks.

Contributions (2/2) — Deeper Analysis

Analyses not present in either baseline

✗ Baseline Gap

GMM used with a single fixed config in Repo 1 — not optimized.



✓ Our Result

Grid search over 32 configs (8×4).
Best: n_components=2, cov='full'.
Result: 98.08% detection without any fault labels.

✗ Baseline Gap

No feature importance analysis in either repository.



✓ Our Result

Ablation study: removed each feature one-by-one. Ia has largest impact (−0.17%). Ib removal slightly improves (+0.04%) — correlated features.

✗ Baseline Gap

Single-dataset evaluation — no cross-repository test.



✓ Our Result

RF trained on Dataset 1 tested on Dataset 2 binary
→ 99.94% accuracy. This suggests strong normal/fault pattern consistency across the two repository datasets.

Results (1/2) — Fault Detection

Binary: Normal vs. Fault | All values from notebook output

Baseline (Repo 1 reproduced)

SVM (RBF)

~99.8% → 99.58%

Decision Tree

~99.8% → 99.58%

Random Forest

~99.8% → 99.63%

MLP

~98.0% → 99.5%

GMM

~99.5% → 98.08%

✓ Classical supervised baselines are close to reported values; MLP and GMM differ due to configuration and threshold changes.

Our Method — 8 Models + Ensemble

🏆 Voting Ensemble

99.75%

Gradient Boosting

99.67%

KNN (k=5)

99.67%

Random Forest

99.63%

SVM (RBF)

99.58%

Decision Tree

99.58%

MLP (ReLU+Adam)

99.50%

GMM (unsupervised)

98.08%

Naive Bayes

97.42%

Logistic Regression

71.30%

Results (2/2) — Fault Classification

6-class: No Fault / LG / LL / LLG / LLL / LLLG | Notebook output

Baseline (Repo 2 reproduced)

SVM

(on clean repo data: ~99%)

Decision Tree

(on clean repo data: ~99%)

Random Forest

(on clean repo data: ~99%)

MLP

(on clean repo data: ~98%)

⚠ Our data: 7,861 valid samples
LLL/LLG confusion: LLL (F1=0.59) & LLLG (F1=0.59)
→ Three-phase faults have similar signatures

Our Method — 8 Models + Ensemble

🏆 Random Forest

88.24%

Voting Ensemble

87.48%

Decision Tree

85.76%

MLP (ReLU+Adam)

84.23%

Gradient Boosting

84.04%

SVM (RBF)

83.98%

Naive Bayes

79.40%

KNN (k=5)

79.28%

Logistic Regression

34.27%

Analysis & Discussion

Why do the results look this way?

Why is detection (~99.75%) >> classification (~88%)?

Detection is binary — the model only needs to separate fault from normal current patterns. Classification must distinguish 6 fault types with similar 3-phase signatures. Confusion matrix confirms: LLL and LLLG are most confused (F1=0.59 each), as both are three-phase events.

What does the ablation study really show?

Removing I_a drops accuracy by 0.17% — the largest single drop. Removing I_b slightly improves (+0.04%), likely due to feature correlation. Result: current features matter, but all 6 features contribute to the ~99.6% baseline.

Why does Ensemble beat individual models?

RF, SVM, GB learn via different mechanisms (bagging vs. margin vs. boosting). Their error patterns differ → soft-vote averaging cancels mistakes. Ensemble: 99.75% detection vs. best single model 99.67%.

Why is GMM useful at 98.08% with no fault labels?

GMM trained only on normal samples forms tight clusters in 6D feature space. Fault conditions fall outside these clusters → threshold-based detection. Best config: $n=2$, $cov='full'$. Practical for field deployment where fault data is scarce.

Conclusions & Future Work

✓ Key Findings



Best Detection

Voting Ensemble (RF+SVM+GB): 99.75%



Best Classification

Random Forest: 88.24%



GMM Unsupervised

98.08% detection — zero fault labels needed



PCA: 4 components

→ 96.9% variance with 33% dimension reduction



LLL / LLLG

F1=0.59 — three-phase faults are hardest to classify



Cross-repo Check

99.94% — strong consistency across repository datasets



Future Work

Fault localization

Estimate distance on line

LSTM / Transformer

On raw time-series data

Address LLL/LLLG

Confusion with richer features

Real field data

Beyond Simulink simulation

IEEE benchmark

14/30-bus validation